Homework 3
Object Detection

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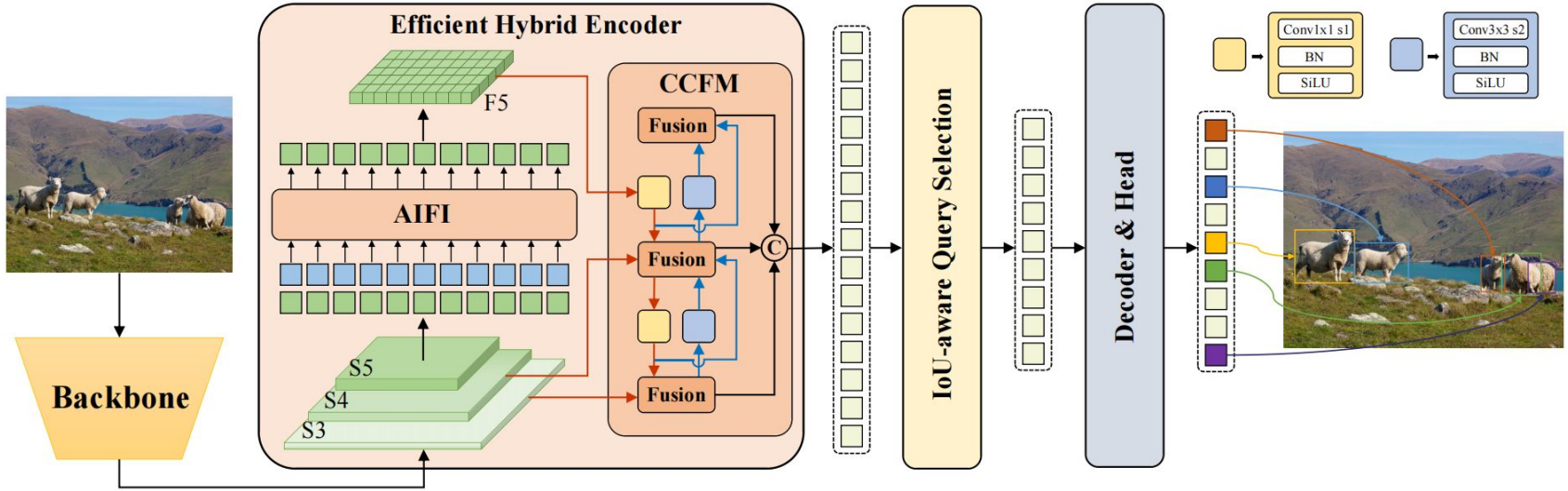
Introduction

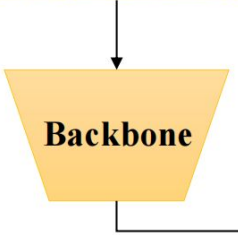
- Train a neural network to do detection on our own dataset
- Practice how to modify the target module in an existing model and implement it as a custom module.
- Model : object detection algorithms
  - RT-DETR
    - You should use **rdetr-resnet18** as your backbone model
    - if you choose another model as your backbone, you will get **0** points in HW3
  - The pretrained model weights are available

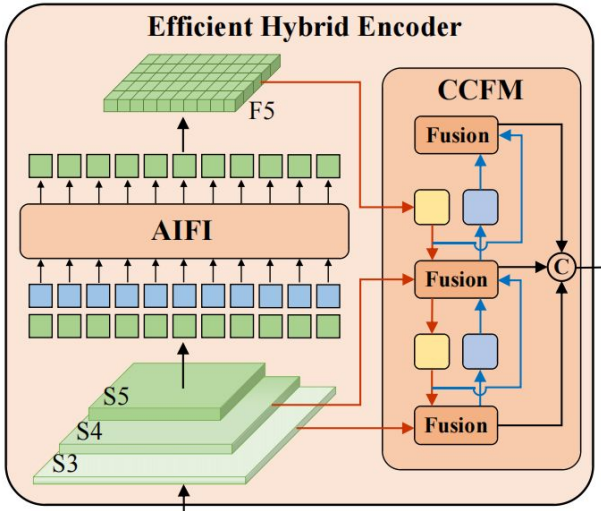
RT-DETR (Real-Time DEtection Transformer)

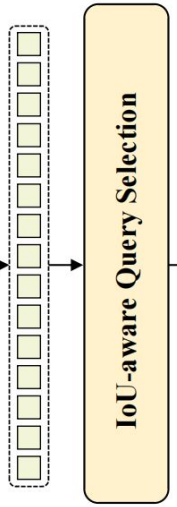


Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., ... & Chen, J. (2024). Detrs beat yolos on real-time object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16965-16974).

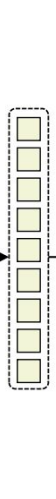
RT-DETR

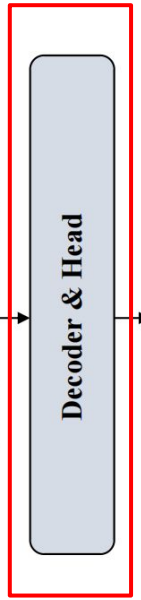

Backbone

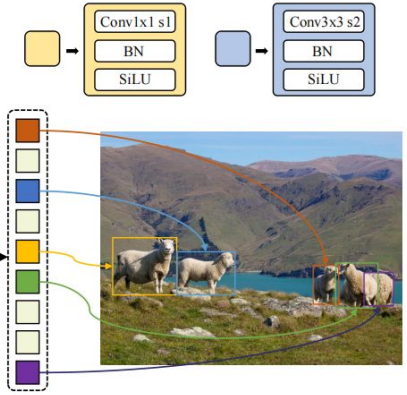

Efficient Hybrid Encoder
F5
AIFI
S5
S4
S3
CCFM
Fusion
Fusion
Fusion
C



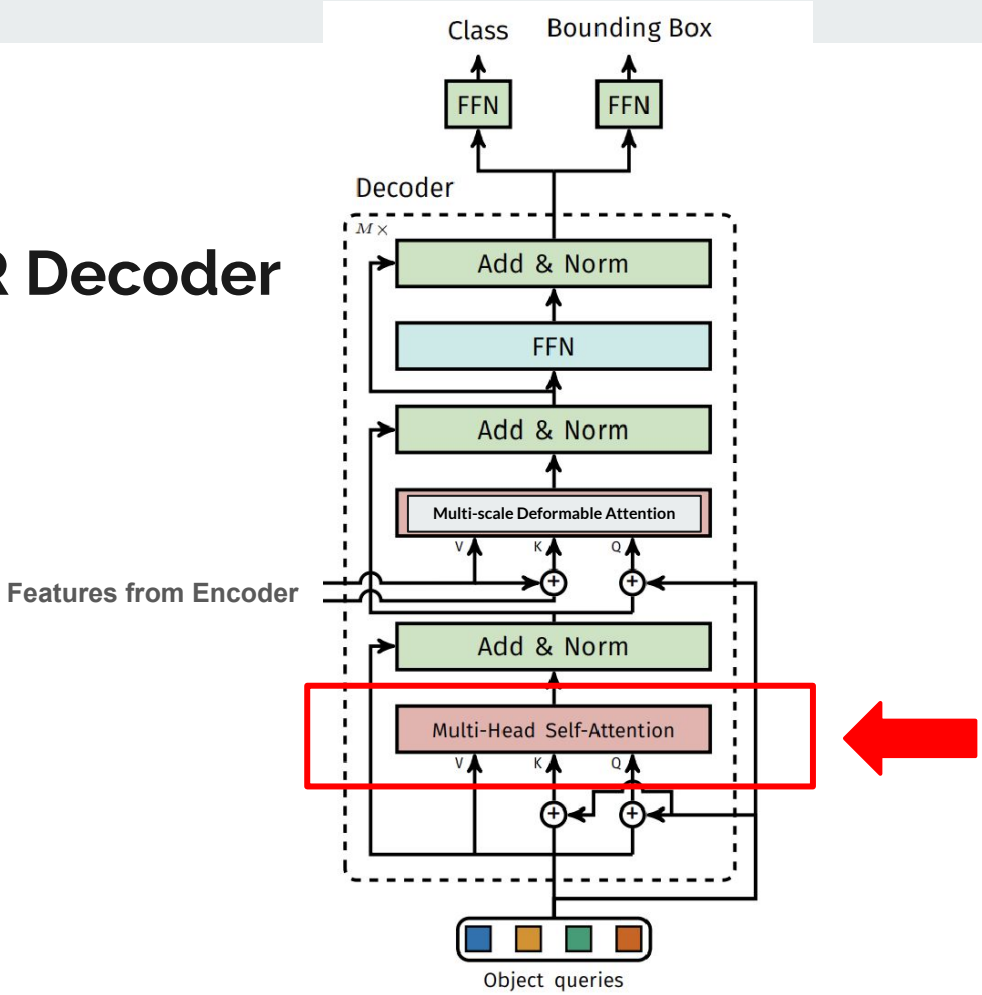
IoU-aware Query Selection




Decoder & Head

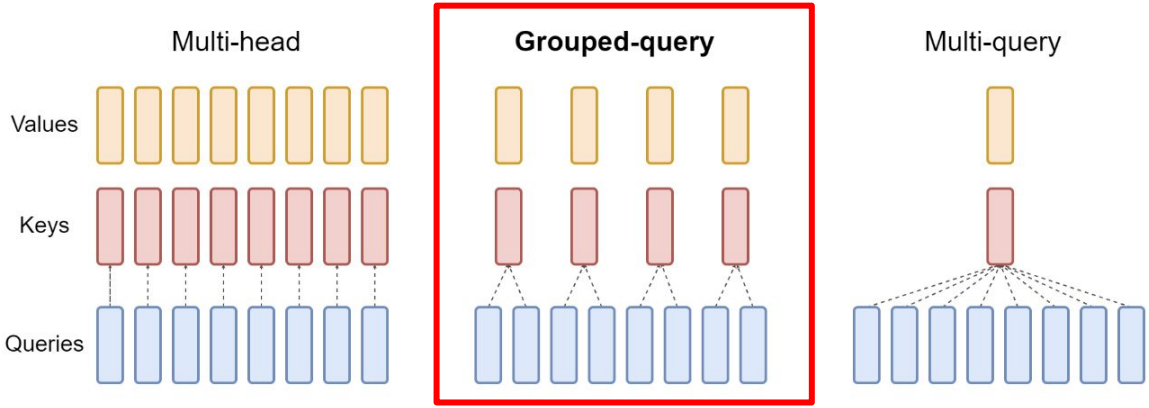

Conv1x1 s1
BN
SiLU
Conv3x3 s2
BN
SiLU

RT-DETR Decoder



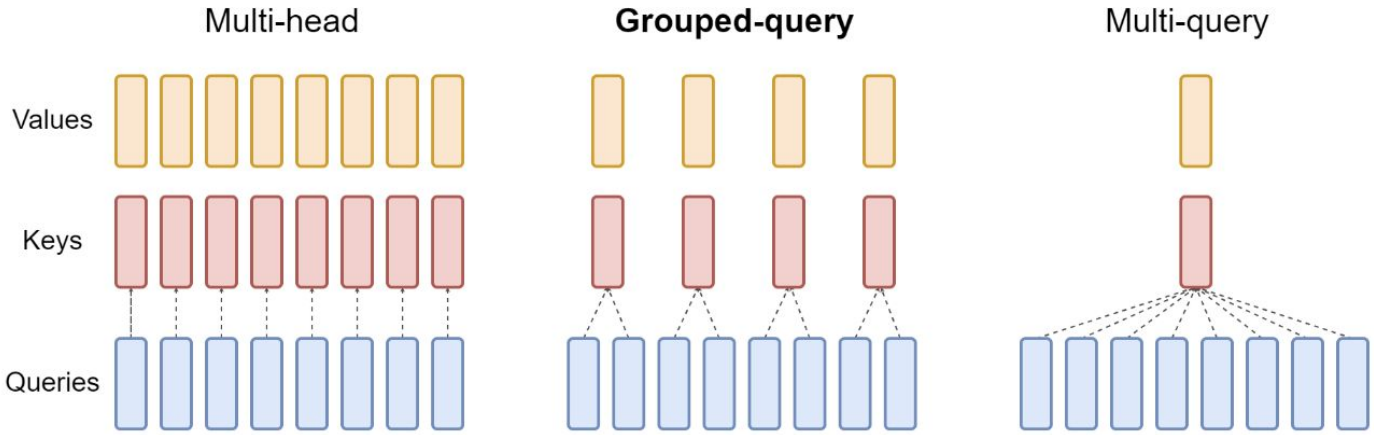
Grouped-query Attention (GQA)

- Several Q heads share the same K and V head
  - Faster than MHSA
  - Keeps almost all the quality of MHSA



Grouped-query Attention (GQA)

- **In this HW, the default number of multi-head (#Q heads) = 8**



 # Q head = 8

 # K, V head = 4
Group_size = 2 (4x2=8)

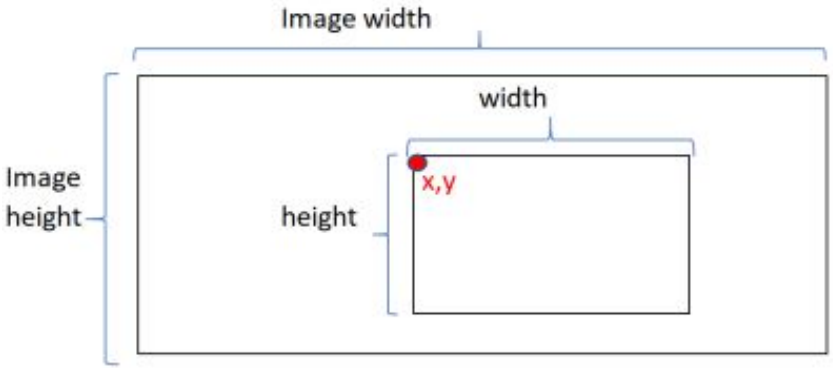
 # K, V head = 1
Group_size = 8 (1x8=8)

Dataset

- GTA video dataset
- You only need to detect car
    - **Only one class**
- 1596 training images, labels
- 227 validation images, labels
- 456 testing images
- Dataset Link (zip file contains train / vaidation/ test split)

Labels

- ./hw3_dataset/{train, val}/annotations.json
- Each bbox is [x_min, y_min, width, height] format (use category_id: 0 to represent car)




```
"annotations": [
    {
        "id": 1,
        "image_id": 1,
        "category_id": 0,
        "bbox": [
            884.0,    x_min
            211.0,    y_min
            34.0,     width
            37.0      height
        ],
        "area": 1258.0,
        "iscrowd": 0,
        "segmentation": []
    },
    {
        "id": 2,
        "image_id": 1,
        "category_id": 0,
        "bbox": [
            1505.0,   x_min
            434.0,    y_min
            181.0,    width
            58.0      height
        ],
```

Evaluation Metrics: mAP (mean Average Precision)

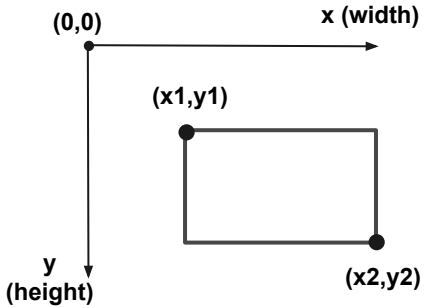
- Most common metric for object detection
- In this HW, mAP defined in the PASCAL VOC 2012 competition is used
- We will use the following github repo to calculate your score

  https://github.com/rafaelpadilla/Object-Detection-Metrics

- It also contains some explanations about how to calculate it
- We will set IoU threshold greater than 0.85 to calculate the testing score

  E.g. python pascalvoc.py -t 0.85 -gtformat xyrb -detformat xyrb -np

Hand in Rules (1/3)



- You should hand in your result by detecting the testing data through your model
- Format **[class confidence left top right bottom]** in pixel wise (1920 x 1080)

14.txt

```
0 0.222222 676 215 770 357
0 0.111111 1345 443 1556 584
0 0.999999 753 55 815 118
```

Hand in Rules (2/3)

- Store each detection result in **[image_name].txt**
- You should have a RT-DETR-Resnet18 model: with the Group-query Attention (GQA)
  - **The number of KV head in your GQA model must < 8**
- Use this model to detect on the testing set
- Submit results in the folder
- You should hand in
  - Result folder **contain testing 456 results**
  - Folder should be **named: result**

Hand in Rules (3/3)

- Your submission should contain
  1. Result folder contain testing 456 results
  2. Report (in pdf)
  3. Code folder **(include your environment and RT-DETR-R18 with GQA checkpoint)**
     - Do not contain dataset and pretrained model weights
     - Please submit the code that can generate the prediction results
- Compress them into **one zip** file name HW3_[studentID].zip

Grading Policy (1/2)

- **(60 points)** GQA implementation on testing set
  - **(20 points)** Replace MHSA with Group-query Attention (GQA) in ALL decoder layers on your model
  - **(40 points)** **Pass the baseline** (**mAP >= 0.90**) by **using the** code **we provide (set IoU threshold to 0.85) to evaluate on the testing set**
- **(10 points)** The Number of model parameters with GQA
  - Rank with your classmates, **10 points in the first half, 5 points in the second half**
  - Your model **must satisfy the following 2 conditions** to join ranking**,** otherwise you will get **0 point** in this part
    1. Use GQA and the number of model parameters **<= 20M**
    2. Has at least mAP (IoU threshold = 0.85) **>= 0.9** on validation set

Way to Get the Number of Parameters

- TA will also use this code to measure your model

```
def num_parameters(model):
    return sum(p.numel() for p in model.parameters())
```

Grading Policy (2/2)

- **(30 points)** Report
  - **(10 points)** Experiment Setup (Data pre-process, Hyperparameters, Training Strategy)
  - **(10 points)** Brief explain your code
    - How do you implement GQA?
    - Training / inference command line
  - **(10 points)** Experiment
    - Validation result screenshot
    - Comparision differernt number of KV head / group size or other method you use
      - Record **validation result** and **#parameters**
- If you used code from GitHub, provide reference in report

Penalty

- Format penalty - **5 points**
  - Submit the result in the wrong name, format, etc.
  - Submit the report not in pdf format
- No validation results are shown in the report - **10 points**
- Deadline: **2025/11/26 23:59**
- Late penalty - **20% per day**
  - 1 day => 80%, 2 day => 60%...
- **You can use any code/pretrained-weight from Github/Huggingface, but don't copy from your classmate!**

Reference

- **RT-DETR:**
  - https://arxiv.org/abs/2304.08069
  - https://huggingface.co/docs/transformers/model_doc/rt_detr
- **Group-query Attention:** https://arxiv.org/pdf/2305.13245
- **Object-Detection-Metrics:** https://github.com/rafaelpadilla/Object-Detection-Metrics

Note

- If you have any questions, please contact us via the E3 system and include **ALL** TAs in your message